I Bharath Chandra Nallani declare that I have completed this assignment completely and entirely on my own, without any consultation with others. I understand that any breach of the UAB Academic Honor Code may result in severe penalties.

Username: bnallaniweb

Database name: bnallani

Password: 1234

Date: Feb 14, 2025

Signature: Bharath Chandra Nallani

1.

```
bnallani=> CREATE TABLE Part_Structure ( MAJOR_P VARCHAR(10) NOT NULL, MINOR_P VARCHAR(10) NOT NULL, QTY INT NOT NULL, PRIMARY KEY (MAJOR_P, MINOR_P)); INSERT INTO Part_Structure (MAJOR_P,
MINOR_P, QTY) VALUES ('P1', 'P3', 2), ('P1', 'P2', 4), ('P2', 'P4', 1), ('P2', 'P3', 3), ('P3', 'P5', 9), ('P4', 'P5', 8), ('P5', 'P6', 3);
CREATE TABLE
INSERT 0 7
```

```
bnallani=> select * from Part_Structure;
 major_p | minor_p | qty
---------+---------+-----
 P1      | P3      |   2
 P1      | P2      |   4
 P2      | P4      |   1
 P2      | P3      |   3
 P3      | P5      |   9
 P4      | P5      |   8
 P5      | P6      |   3
(7 rows)

bnallani=>
```

2.

```
bnallani@cs-vulcan-2:~$ javac Questioncode2.java
bnallani@cs-vulcan-2:~$ java Questioncode2 bnallani bnallaniweb 1234
Subparts of Part P1: P1 P2 P3 P5 P6 P4 P5 P6 P3 P5 P6
bnallani@cs-vulcan-2:~$
```

3.

```
bnallani=> CREATE TABLE Part_Structure1 ( MAJOR_P VARCHAR(10) NOT NULL, MINOR_P
VARCHAR(10) NOT NULL, QTY INT NOT NULL, PRIMARY KEY (MAJOR_P, MINOR_P)); INSERT
INTO Part_Structure1 (MAJOR_P, MINOR_P, QTY) VALUES ('P1', 'P3', 2), ('P1', 'P2'
, 4), ('P2', 'P4', 1), ('P2', 'P3', 3), ('P3', 'P5', 9), ('P4', 'P7', 8),('P5',
'P6', 3);
```

```
bnallani=> select * from Part_Structure1;
 major_p | minor_p | qty
---------+---------+-----
 P1      | P3      |   2
 P1      | P2      |   4
 P2      | P4      |   1
 P2      | P3      |   3
 P3      | P5      |   9
 P4      | P7      |   8
 P5      | P6      |   3
(7 rows)
```

```
bnallani=> CREATE TABLE Part_Structure1 ( MAJOR_P VARCHAR(10) NOT NULL, MINOR_P VARCHAR(10) NOT NULL, QTY INT NOT NULL, PRIMARY KEY (MAJOR_P, MINOR_P)); INSERT INTO Part_Structure1 (MAJOR_P
, MINOR_P, QTY) VALUES ('P1', 'P3', 2), ('P1', 'P2', 4), ('P2', 'P4', 1), ('P2', 'P3', 3), ('P3', 'P5', 9), ('P4', 'P7', 8),('P5', 'P6', 3);
CREATE TABLE
INSERT 0 7
```

```
bnallani@cs-vulcan-2:~$ javac Questioncode3.java
bnallani@cs-vulcan-2:~$ java Questioncode3 bnallani bnallaniweb 1234
P6 378
P7 32
bnallani@cs-vulcan-2:~$
```

4. A.

```
bnallani=> CREATE OR REPLACE FUNCTION Traversal(upper_p character varying) RETURNS VOID AS $$ DECL
ARE lower_p character varying; BEGIN RAISE NOTICE '%', upper_p; FOR lower_p IN SELECT minor_p FROM
 Part_Structure1 WHERE major_p = upper_p ORDER BY minor_p LOOP PERFORM Traversal (lower_p); END LO
OP; END; $$ LANGUAGE plpgsql;
```

```
bnallani=> CREATE OR REPLACE FUNCTION Traversal(upper_p character varying) RETURNS VOID AS $$ DECLARE lower_p character varying; BEGIN RAISE NOTICE '%', upper_p; FOR lower_p IN SELECT minor
_p FROM Part_Structure1 WHERE major_p = upper_p ORDER BY minor_p LOOP PERFORM Traversal (lower_p); END LOOP; END; $$ LANGUAGE plpgsql;
CREATE FUNCTION
bnallani=> SELECT Traversal('P1');
NOTICE:  P1
NOTICE:  P2
NOTICE:  P3
NOTICE:  P5
NOTICE:  P6
NOTICE:  P4
NOTICE:  P7
NOTICE:  P3
NOTICE:  P5
NOTICE:  P6
 traversal
-----------
```

B.

```
bnallani=> CREATE OR REPLACE FUNCTION leafpart(upper_p character varying, sbrt character varying) RETURNS INTEGER AS $$ DECLARE crt_p RECORD; sbrt_ct INTEGER := 0; BEGIN FOR crt_p IN SELECT
 minor_p, qty FROM Part_Structure1 WHERE major_p = upper_p ORDER BY minor_p LOOP IF crt_p.minor_p = sbrt THEN sbrt_ct := sbrt_ct + crt_p.qty; ELSE sbrt_ct := sbrt_ct + crt_p.qty * leafpart(
crt_p.minor_p, sbrt); END IF; END LOOP; RETURN sbrt_ct; END; $$ LANGUAGE plpgsql;
CREATE FUNCTION
bnallani=> SELECT leafpart('P1', 'P6') AS "P6", leafpart('P1', 'P7') AS "P7";
 P6  | P7
-----+----
 378 | 32
(1 row)

bnallani=>
```

5.

```
bnallani=> \q
bnallani@cs-vulcan-2:~$ javac Questioncode5.java
bnallani@cs-vulcan-2:~$ java Questioncode5 bnallani bnallaniweb 1234
Connected to database: bnallani
Select an option:
1. Execute SQL query
2. Insert data
3. Update data
4. Delete data
5. Create table
6. Exit
```

```
Select an option:
1. Execute SQL query
2. Insert data
3. Update data
4. Delete data
5. Create table
6. Exit
2
Enter your SQL insert query:
INSERT INTO users (id, name) VALUES (1, 'John');
```

```
Select an option:
1. Execute SQL query
2. Insert data
3. Update data
4. Delete data
5. Create table
6. Exit
3
Enter your SQL update query:
UPDATE users SET name = 'Johnny' WHERE id = 1;

1 rows updated.
```

```
Select an option:
1. Execute SQL query
2. Insert data
3. Update data
4. Delete data
5. Create table
6. Exit
4
Enter your SQL delete query:
DELETE FROM users WHERE id = 1;
1 rows deleted.
```

```
Select an option:
1. Execute SQL query
2. Insert data
3. Update data
4. Delete data
5. Create table
6. Exit
6
Exiting...
bnallani@cs-vulcan-3:~$
```

6.

```
> CREATE TABLE Students1(snumber INT,  sname VARCHAR(100), DoB DATE);
```

```
bnallani=> select *from Students1;
 snumber |     sname     |     dob
---------+---------------+------------
 101     | John Doe      | 2000-05-12
 102     | Johnny Depp   | 1995-08-20
 103     | Alice Smith   | 1998-03-10
 104     | Bob Johnson   | 2001-07-25
 105     | Emily Davis   | 1999-11-30
(5 rows)
```

```
bnallani=> INSERT INTO Students1 (Snumber, Sname, DoB) VALUES ('101', 'John Doe'
, '2000-05-12'), ('102', 'Johnny Depp', '1995-08-20'), ('103', 'Alice Smith', '1
998-03-10'), ('104', 'Bob Johnson', '2001-07-25'), ('105', 'Emily Davis', '1999-
11-30');
```

```
bnallani@cs-vulcan-2:~$ javac Questioncode6.java
bnallani@cs-vulcan-2:~$ java Questioncode6 bnallani bnallaniweb 1234
Student Name: John Doe
Student Number: 101
Date of Birth (YYYY-MM-DD): 2000-05-12

--- Student Records ---
Number: 101, Name: John Doe, DoB: 2000-05-12
bnallani@cs-vulcan-2:~$ java Questioncode6 bnallani bnallaniweb 1234
Student Name: jeff
Student Number: 222
Date of Birth (YYYY-MM-DD): 1899-05-21

--- Student Records ---
No matching records found.
bnallani@cs-vulcan-2:~$
```